Exploring Trust in AI

A MULTILEVEL ANALYSIS OF INFLUENCING FACTORS, TRUST EROSION AND TRUST CALIBRATION STRATEGIES

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THE PROBLEM

AI use is high. Checking is not.

66% use AI regularly

46% are willing to trust AI

39% have had any AI training

Two in three employees rely on AI output without checking its accuracy.

56% of employees have made mistakes at work because of AI.

48,340 people, 47 countries · Gillespie et al. (2025), KPMG & Univ. of Melbourne

OUR GOAL

Calibrated trust, not more trust

Under-trust
Rejects useful tools and loses their benefits

Calibrated trust
Relies on AI where it is reliable, questions it where it is not

Over-trust
Accepts wrong outputs without checking

Calibration framing: Jacovi et al. (2021), ACM FAccT

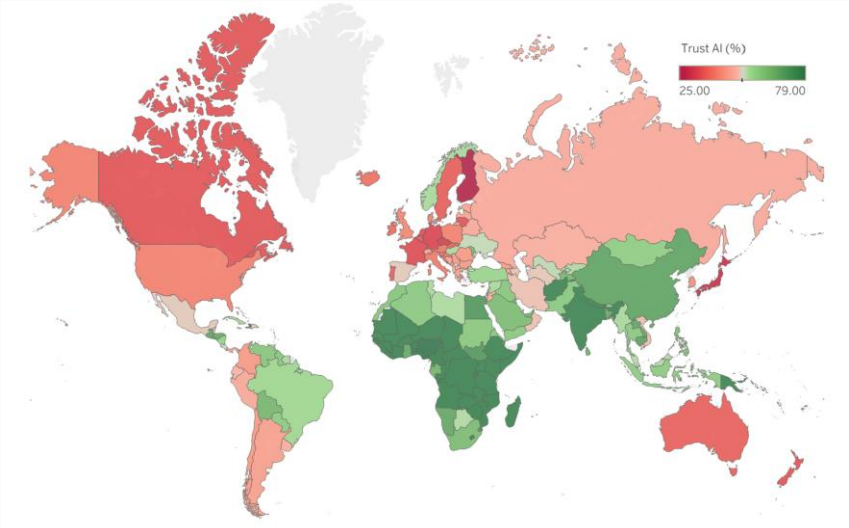
THREE LEVELS OF ANALYSIS, THREE DATASETS

01 · COUNTRY LEVEL
Which national traits predict AI trust?
47 labelled + 146 unlabelled countries · 7 indicators · 9 regression models

02 · INDIVIDUAL INTERACTIONS
What moves a user away from blind trust?
1,000 AI interaction records · 4 trust profiles · multinomial logistic regression

03 · AFTER AI ERRORS
Who loses trust after an error, and does it return?
301 AI use sessions (201 with errors) · ANOVA · K-means clustering

01 Country level: national traits predict AI trust, but only partly



Share of people trusting AI: survey data for 47 countries plus KNN predictions for 146 more (193 in total). Many developing countries are predicted to trust AI more than developed ones.

KNN
k = 3
Best Model

0.26
R^2

8.89
RMSE

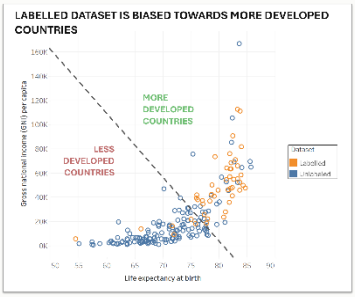
MOST INFLUENTIAL PREDICTORS

01 **Age dependency ratio**
02 Mean years of schooling
03 Government effectiveness
04 Internet usage
05 GNI per capita

FINDINGS

- Age structure may reflect generational differences in familiarity with technology
- Education level and internet usage may reflect digital literacy and exposure to AI
- Country development tend to have greater technological advancements which affect AI trust

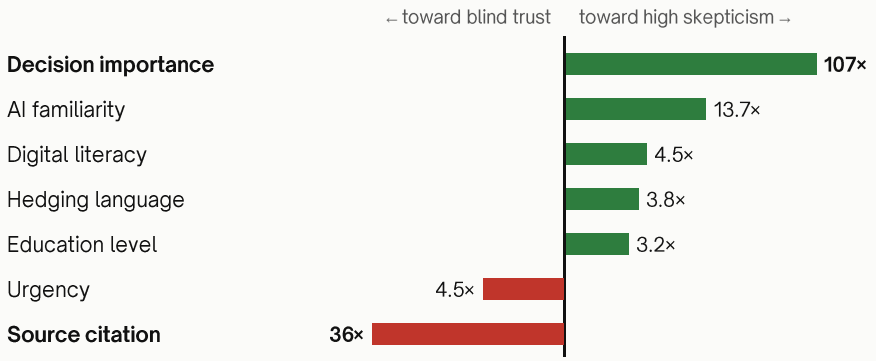
LIMITATION



Only 47 labelled, mostly developed countries, so predictions for lower-income countries are less reliable. Associations, not causes.

02 Individual interactions: high stakes and expertise breed scrutiny

ODDS OF HIGHLY SKEPTICAL VS BLIND TRUST



Multinomial logistic regression, 1,000 records, Blind Trust as reference; significant odds ratios ($p < 0.05$), pseudo $R^2 = 0.48$. Only 57 users were Highly Skeptical, so treat this group as indicative.

TRUST PROFILES (N = 1,000)



Blind trust 36.7% | Moderate 31.1%
Skeptical 26.5% | Highly skeptical 5.7%

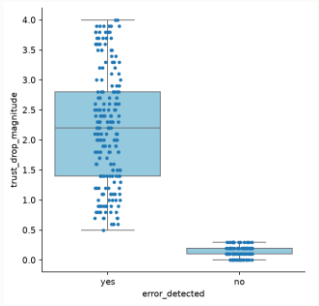
Response quality: citations calm skeptics; hedging raises doubt.

Query context: high stakes trigger scrutiny; time pressure suppresses it.

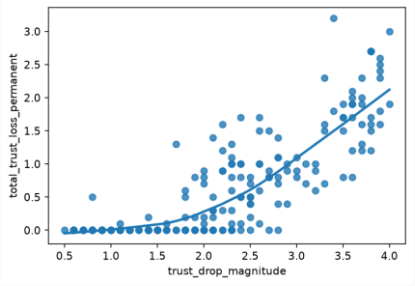
User characteristics: experts are more critical, and answers that match a user's beliefs make them less likely to be skeptical.

The same answer can earn blind trust or deep skepticism, depending on who asks and why.

03 After AI errors: detected errors cut trust, and large drops last



Trust drop after error varies widely



Drops below about 1.5 tend to recover fully; larger drops leave permanent loss

USER TYPES · AVG TRUST DROP

Uncritical Enthusiasts 3.32
Open to tech, low critical thinking

Disengaged Users 1.81
Low openness and critical thinking; little trust to lose

AI-Literate 1.64
Open to tech and think critically

PROPOSED SOLUTIONS

Building AI literacy and accountability

1. **Teach AI verification** to students and Uncritical Enthusiasts.
   e.g. Common Sense Education's Online Learning, *How to SIFT with AI.*
2. **Bring AI familiarity** to disengaged and older groups.
   e.g. IMDA GenAI workshops for 6,000+ seniors
3. **Assurance and accountability** for AI companies.
   e.g. AI Verify (2022); Moffatt v. Air Canada (2024)

Designing AI that signals when to trust it

4. **Real, checkable sources**, since citations drive trust.
   e.g. Mata v. Avianca (2023): invented cases
5. **Reduce sycophancy**, since agreement lowers scrutiny.
   e.g. OpenAI's GPT-4o update (2025)
6. **Signal limits**, especially in high-stakes areas.
   e.g. Google limiting AI Overviews on health (2024)

Calibrated trust is built by users, institutions and AI developers together.

Limitations: Observational data, small samples, some possibly synthetic. Findings are associations, not causes.

Next: Test these proposals in controlled experiments that record trust, AI accuracy and checking together.

Data: Mind Lab (2025), Global AI trust and attitudes by country; UNDP Human Development Index; World Bank World Development Indicators; Fatima (2025), AI trust insights; Landge, Human trust decay in AI. **Sources:** Gillespie et al. (2025), KPMG & Univ. of Melbourne; Jacovi et al. (2021); Common Sense Education; Koh, The Straits Times (2025); PDPC (2022); Moffatt v. Air Canada, 2024 BCCRT 149; Mata v. Avianca (S.D.N.Y. 2023); OpenAI (2025); Forbes (2024).